



Thermal runaway propagation behavior of the Cell-to-Pack battery system

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ABSTRACT

Structurally compact battery packs significantly improve the driving range of electric vehicles. Technologies like Cell-to-Pack increase energy density by 15%–20%. However, the safety implications of multiple tightly-packed battery cells still require in-depth research. This paper studies thermal runaway propagation behavior in a Cell-to-Pack system and assesses propagation speed relative to other systems. The investigation includes temperature response, extent of battery damage, pack structure deformation, chemical analysis of debris, and other considerations. Results suggest three typical patterns for the thermal runaway propagation process: ordered, disordered, and synchronous. The synchronous propagation pattern displayed the most severe damage, indicating energy release is the largest under the synchronous pattern. This study identifies battery deformation patterns, chemical characteristics of debris, and other observed factors that can both be applied to identify the cause of thermal runaway during accident investigations and help promote safer designs of large battery packs used in large-scale electric energy storage systems.

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1. Introduction

The rise of the new energy vehicle industry is facilitating the global transition from fossil energy to renewable energy [1–3]. The cumulative sales of new energy vehicles exceed 6.5 million by 2021. Among the new energy vehicles, the proportion of pure electric vehicles (PEVs) is as high as 73% [4]. However, consumers are still worrying about the electric driving range of PEVs, as the energy density of the battery system is not sufficient yet [5].

The electric driving range of the PEVs is determined by the energy stored within the battery pack, which has limited space on board. There are two ways for increasing the energy density of the battery pack of the PEVs. One is to increase the energy density of the lithium-ion batteries (LIBs) [2,6,7]. However, upgrading the cell chemistry into higher level of energy density is rather hard.

Therefore, people choose to take another way to acquire higher energy density on-board, i.e., designing battery pack with more compactness. The engineers realized that the structures for the battery modules might be unnecessary in the battery system, thereby leading to a revolution in the structure design of the battery pack [8–10]. In September 2019, CATL announced the world's first Cell-to-Pack (CTP) system used to massively produce for electric vehicles. The structural components of battery module have been eliminated in the CTP design, resulting in an increase of the energy density of as high as 15%–20%. Moreover, the number of the elements that are used to form a battery pack is reduced by 40%, with 50% increase in the production efficiency [11]. However, the failure behavior of the CTP system remains unclear. As the compactness increases in the battery pack, the hazard caused by failure propagation might be worse as more battery cells were squeezed inside a limited space.

A series of studies on the thermal safety of LIBs, modules, and battery packs have been carried out to improve the safety of electric vehicles [12–16]. At the cell level, the state-of-the-art study

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focuses on revealing the thermal runaway (TR) mechanism underlying the failure characteristics of temperature rise, smoke bursting, fire and explosions, and technologies to improve battery safety [17–20]. Cai et al. [21] investigated the TR mechanism of a commercial $\text{Li}[\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}]\text{O}_2(\text{NCM523})/\text{graphite}$ 18,650 type cylindrical battery with various state of charge (SOCs) through differential scanning calorimetry (DSC) tests on the single and mixed components. Wang et al. [2] investigated the overcharge behaviors for large format LIBs with different cathode materials, and results showed that the $\text{LiFePO}_4(\text{LFP})$ lithium-ion cell had poor tolerance against the overcharge risk. Walker et al. [22] utilized the fractional thermal runaway calorimetry (FTRC) to gauge the influences of cell geometry and triggering method on the TR behaviors of lithium-ion batteries. Henriksen et al. [23] used a 20 L spherical explosive tank to evaluate the explosive hazard of three common electrolytes, i.e., ethyl methyl carbonate (EMC), dimethyl carbonate (DMC), and diethyl carbonate (DEC), at 373 K and 100 kPa, and compared the explosive hazard with that of H_2 , CH_4 , and C_3H_8 . Ouyang et al. [24] investigated the application of a non-flammable electrolyte with a solvent mixture of fluoroethylene carbonate (FEC) and DMC in $\text{Li}(\text{Ni}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1})\text{O}_2$ cathode lithium-ion batteries. The results showed that when the ratio of FEC to DMC was 3:7, the battery exhibited better safety performance in accelerated rate calorimetry (ARC) test, nail penetration test, and overcharge test. The physicochemical characteristics of the vented particles during TR depend on morphology, compound type, element composition etc. [25]. Wang et al. [26] investigated the effect of DMC addition on the combustion characteristics of vent gases from LIBs. For the different components of exhaust gases in LIB, most of the combustion parameters were reduced or remained unchanged after DMC addition. The results indicated that adding DMC can reduce the fire risk of LIB vent gases.

At the level of battery module, the thermal safety research mainly focuses on mechanism of TR propagation, as well as the influence of SOCs, ambient pressure, and triggering methods on the behavior of TR propagation [2,16,27]. Khan et al. [28] developed a mathematical model for speculating TR propagation in a $\text{Li}[\text{Ni}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}]\text{O}_2(\text{NCM811})/\text{C}$ LIB module after fast charging and conducted experimental validation. The effects of charging rate, inter-cell spacing, trigger temperature, velocity, and propagation distance on the TR propagation behavior were investigated. Jia et al. [29] investigated the characteristics of TR propagation for the lithium-ion battery modules at 95, 70, and 35 kPa, and the results indicated that the TR propagation behaviors become weaker as the pressure decreases. Li et al. [30] studied the TR propagation characteristics of battery module under different SOCs, and they found that the propagation time of TR becomes longer as the SOC decreases. Wang et al. [31] investigated the influence of cycling aging, connection mode, arrangement mode, and SOC on the propagation behavior of TR in cylindrical 18,650 LIBs.

There are few studies on the TR propagation characteristics among battery packs [32]. There are many coupling factors in the TR propagation process of battery pack, and the TR behavior is complicated. It is difficult to determine the TR propagation principle of battery pack. The TR propagation between battery modules in pack is driven by solid heat transfer, hot smoke diffusion, and flame radiation. CTP systems enable structural innovation, increase the energy density of battery packs, and promote the scale of electric vehicles. There is no literature on TR propagation of CTP. Hence, it is extremely important to investigate and understand the TR propagation behaviors of CTP battery system.

This study examines the behavior of TR propagation in CTP battery systems through experimentation. During the testing, side surface temperature and volatile organic compounds detector volt-

age were recorded. The TR propagation behaviors, the flame sequence, the vent gas diffusion, and failure propagation mechanisms for the CTP battery system are revealed by in-depth data analysis. The physicochemical properties of smoke particles from TR batteries were analyzed in detail and systematically compared. The results provide suggestions for improving the safety design of the CTP battery system and early warning in the CTP, and can contribute to trace the origin pattern of LIBs fire accident.

2. Experimental

2.1. The Cell-to-Pack battery system

Fig. 1 illustrates the CTP battery system that was set up for the experimental study, including the arrangement of battery cells, cooling channels, and other components. The CTP battery system comprises five large modules with double-row battery cells. The large modules are denoted as M1–M5. Each one has $9 \times 2 = 18$ battery cells connected in series. The electrical connection structure of the battery pack is one parallel and 90 series. There is a U-shaped hard rubber layer between the batteries. It has a density of 1.3 g cm^{-3} , a specific heat capacity of $1700 \text{ J kg}^{-1} \text{ K}^{-1}$, and a thermal conductivity of $0.15 \text{ W m}^{-1} \text{ K}^{-1}$. The M4 and M5 are vertically placed up and down, whereas M1–M3 are horizontally arranged. The battery management system (BMS) and other electrical components are installed in the middle of the CTP system. The vertical distance between M4 and M5 is 3.4 cm. The distance between the three modules (M1, M2, and M3) is 4.2 cm. The battery cell has a capacity of 156 Ah, and the cathode material is NCM523. The geometry of the battery is $148 \text{ mm} \times 79 \text{ mm} \times 103 \text{ mm}$. There are four jelly rolls inside the battery cell. The nominal voltage is of the battery pack 330 V, and the specific energy is 50 kWh. The weight of the battery pack is 320 kg, and the SOC of the CTP system is preset to 100%.

2.2. Lateral heating test to trigger thermal runaway

The TR is firstly triggered at the corner of the M4 module, as shown in Fig. S1. Lateral heating is used to heat the first cell into thermal runaway. The geometry of the heating plate is $148 \text{ mm} \times 103 \text{ mm} \times 3 \text{ mm}$, and the power of the heater is 700 W. An insulation pad is placed between the end plate and the heater to reduce the heat loss through the end plate. The heat conduction between the heating plate and the first-cell was enhanced by spreading conductive glue on the contact surface. The end plate is milled and thinned after the heating plate is added, which maintains the preload of module. High voltage insulation of the battery pack was checked after the heater was inserted into the corner of M4. During the TR propagation test, the heater will keep open until the TR is triggered in the first-cell. After the first-cell in the M4 module is triggered by lateral heating, the CTP system will suffer horizontal and vertical failure propagation, on which we are going to study.

2.3. Sensor location

The failure propagation behavior of the CTP system was observed by placing thermal and gas sensors, of which the locations are shown in Fig. 2. A total of 118 K-type thermocouples, 3 smoke sensors, and 12 volatile organic compounds (VOC) sensors was arranged in the CTP battery system.

The thermocouples are marked in red dots. There are 22 thermocouples placed on the top cover of the CTP system. 8 thermocouples are pasted outside, and 14 thermocouples are inside.

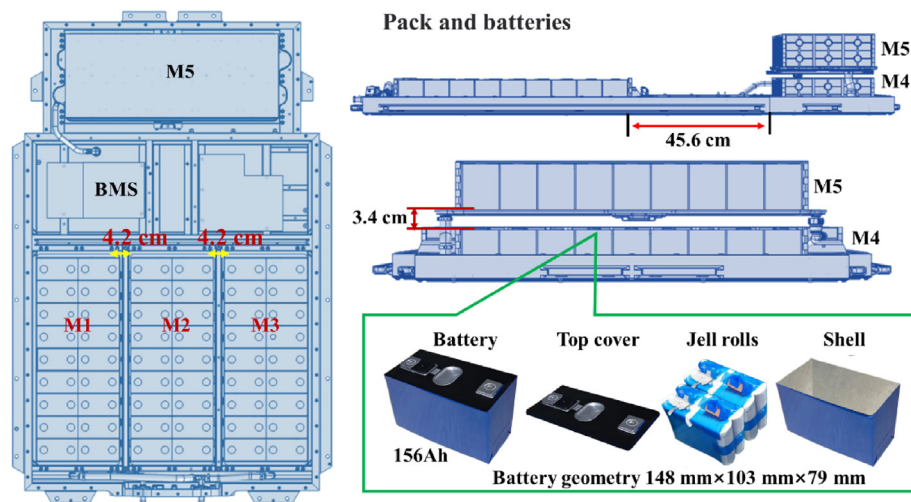


Fig. 1. The specifications of the CTP battery system.

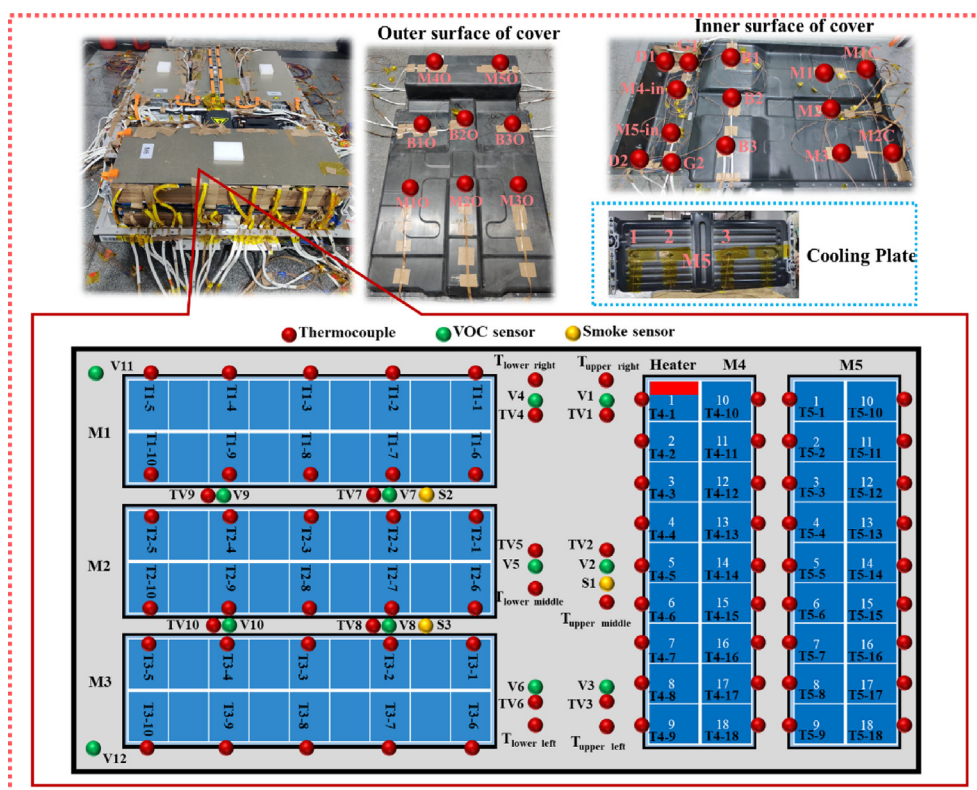


Fig. 2. The sensors location for measuring the failure behavior of the CTP battery system.

There are 66 thermocouples placed on the side of the cells. For the M4 and M5 module which suffer the harshest propagation damages, each cell was allocated one thermocouple to monitor the TR propagation process, and the thermocouples are attached to the side of the cell. The temperature of the cooling plate between M4 and M5 was measured by 4 thermocouples, denoted by liquid plate i ($i = 1, 2, 3, 4$). Cooling plate is attached to the bottom of M5 and therefore not in contact with M4. The thermocouples are placed on the bottom side of the cooling plate and therefore directed towards the vents of M4 cells. For the M1–M3 modules, every second cell has one thermocouple to measure temperature, and the thermocouples are attached to the side of the cell. The thermal

management system is equipped with 6 thermocouples. $T_{\text{upper right}}$ and $T_{\text{lower right}}$ are located on the heater side. $T_{\text{upper middle}}$ and $T_{\text{lower middle}}$ are in the middle. $T_{\text{upper left}}$ and $T_{\text{lower left}}$ are located on the non-heater side.

The smoke sensors (marked in yellow dots) are used to detect the diffusion path of vent gas, The VOC sensors (marked in green dots) are used to detect the diffusion path of the vaporized organic solvent. Most of the smoke and VOC sensors are placed together with thermocouples, and they are represented by TV1 to TV12. To reduce the damage to the connection wire caused by high temperature exposure, the sensor connection wire is treated with high-temperature resistance.

2.4. Video capture

The experiment video is captured by multiple cameras from different angles surrounding the CTP system, as shown in Fig. S2. Three cameras and two infrared thermal imagers are arranged in the experiment. The cameras are placed at different sides towards the CTP system, the venting gas and the burst flame can be recorded. The infrared thermal imager is used to measure the temperature of the burst flame during the experiment.

2.5. Smoke particle characterization

The smoke particles vented from failed cell usually have a diameter of less than 100 μm that can be suspended in the air. The smoke particles can be collected from the inner surfaces of the top cover and from the coverage over the battery poles. The chemical and physical characteristics of the smoke particles help infer the cause of TR. For instances, smoke from candle contains paraffin wax, and smoke from cigarette contains nicotine. The smoke vented from batteries includes critical information on the triggering causes of thermal runaway. Accident investigators can trace the cause of thermal runaway based on the physical and chemical properties of the smoke particles. In this study, X-ray diffraction (XRD), scanning electron microscope (SEM), energy-dispersive X-ray spectroscopy (EDS), X-ray photoelectron spectroscopy (XPS), inductively coupled plasma-mass spectrometry (ICP-MS), and elemental analyzer (CHNSO) were utilized to analyze smoke particles, as shown in Fig. 3.

3. Results and discussion

3.1. Characteristics of thermal runaway propagation

During the experiment, TR initiated from one cell at the corner of the M4 module, and both the M4 and M5 modules experienced TR propagation. There was no TR propagation among the M1, M2, and M3 modules. The large gap between the M4 and M5 modules and the M1–M3 modules hinders the heat transfer for TR propagation. Therefore, further discussions mainly focus on the phenomenon that occurred on M4 and M5.

3.1.1. TR propagation time and temperature response

Fig. 4 shows the temperature curves of the cells in M4 and M5 during TR propagation test. There are three propagation patterns,

i.e., the ordered propagation, the disordered propagation, and the synchronous propagation. Ordered propagation is defined as the occurrence of TR in each individual cell within a module in the estimated direction of heat transfer, with a predetermined order that is consistent with the TR propagation sequence within the M4 module, i.e., propagating from the side near the heater to the end of the module. Synchronous propagation refers to the occurrence of TR in two or more cells within the module at almost the same time (time difference less than 10 s). Disordered propagation refers to the opposite or random propagation order of individual cells within the module compared to sequential propagation. When the ordered propagation is speeding up, the synchronous propagation is probably to come up.

According to Fig. 4(a), the batteries with ordered propagation in the M4 module started from cell 1# to 6#, the total propagation time of the 6 cells is 1787 s, with an average time of 357 s. After the ordered propagation, the total propagated time of the remaining 12 cells of 7#–18# is only 567 s, with an average propagation time of 54 s. Among them, the propagation time of 10#–14# cells is the shortest, and the total time interval is only 7 s, indicating the intensive energy release process for the synchronous propagation. The synchronous propagation corresponds to a larger heat release rate, which is more dangerous than the ordered propagation.

Fig. 4(b) shows the temperature curve of the M5 module that is located on the top of the M4 module, and the corresponding propagation duration is highlighted by the red area in Fig. 4(b). Fig. 4(c) is a magnified image of TR propagation of M5 module. The propagation process includes synchronous propagation and disordered propagation. The upper module (M5) was significantly preheated by the lower module (M4). Therefore when the TR was triggered in M5, the propagation speed was very fast. All the 18 cells in the M5 module underwent TR within 10 min. The first cell in the M5 module that had TR is the 5# cell, located at the middle of the module. The second cell that underwent TR in the M5 module is the cell 10#, 290 s after cell 5# triggers TR. The propagation order displayed a “disordered propagation” at later stage of TR propagation in a CTP system. In the M5 module, the TR propagation of the other 17 cells (except 5#) took a total time of 312 s, with an average spreading time of 18 s. The TR of cells 3# and 4# in M5 even occurred at the same time (3500 s after the heating started). The fast propagation for the whole M5 module complied with the synchronous propagation pattern.

During the ordered TR propagation, the temperature at the side surface of the cell is approximately 559 $^{\circ}\text{C}$. When the synchronous

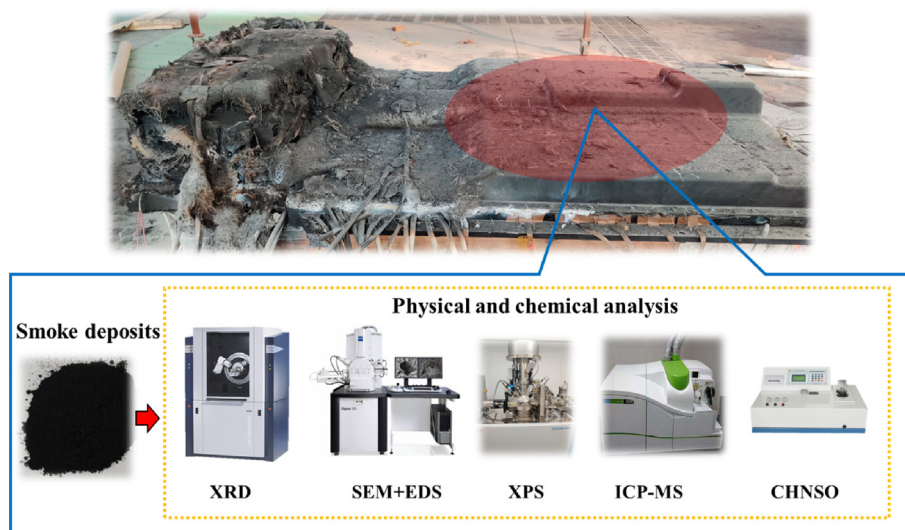


Fig. 3. Instruments used for the chemical analysis of the smoke particles.

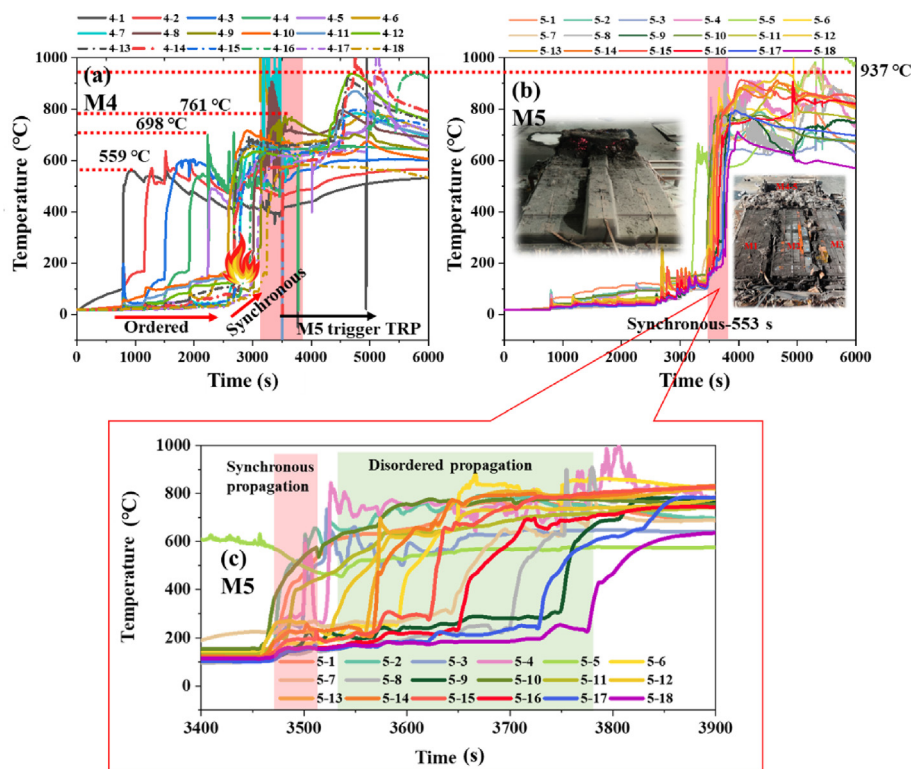


Fig. 4. The thermal runaway propagation characteristics of the Cell-to-Pack system.

TR propagation occurs, the temperature at the side surface of the cell rises to 698 °C. The synchronous propagation was also accompanied by fierce jet fire that was observed outside the cover of the battery pack. The temperature at the side surface of the battery cell further rose to higher level, i.e., 937 °C for the M5 module that fully underwent synchronous TR propagation.

3.1.2. The sequence of thermal runaway propagation

Fig. 5 concludes the propagation sequence of the 36 cells in M4 and M5, according to the temperature profile and the video records. # indicates the battery number. The circles numbered 1–35 indicate the TR time sequence. Same number on the battery indicates that both batteries trigger TR at the same time.

The cells 1#–6# displayed a typical “ordered propagation”, with an average propagation time of 297 s. The “synchronous propagation and disordered propagation” soon occurred when the 11# cell in the M4 module underwent TR as the 7th failing cell. Within 7 s, the 10#–14# cells (highlighted in purple in Fig. 5) in the M4 module underwent TR nearly together, with an order of “11#–10#–14#–12#–13#”. The propagation order displayed the feature of “disordered propagation”; whereas the fast spreading speed inferred that the propagation here is the “synchronous propagation”. “Disordered propagation” followed to come in the M4 module, and the further propagation sequence is “15#(12th)–7#(13th)–16#(14th)–8#(15th)–17#(16th)–9#(17th)–18#(19th)”, with an average propagation time of 69 s per cell. The energy released by the M4 module can torch the upper module M5 to trigger TR propagation. Therefore, the 18th cell that underwent TR in the M4 and M5 modules was the 5# cell in the upper M5 module. In summary, the TR propagation law for the M4 module includes: (1) a stage of ordered propagation for cells 1#–6#, with an average propagation speed of 297 s per cell, (2) a stage of synchronous and disordered propagation for cells 10#–14#, with an average propagation speed of 7 s per cell, and (3) a stage

of disordered propagation for cells 7#–9# and 15#–18#, with an average propagation speed of 69 s per cell.

The TR started to be triggered in M5 when the M4 module was close to the end of TR propagation. The 5# cell (18th) in M5 was the first cell to trigger TR, followed by the 10#(20th), 2#(21st), 1#(22nd), and 11#(23rd) cell near the left end. The 3# and 4# (both ranked the 24th) cells underwent TR at the same time. The remaining 11 cells ran into TR rapidly and disorderly within 248 s. The total TR propagation time for the M5 module was 553 s, with an average propagation time of 30.7 s. The propagation speed in the M5 module is much faster than that in the M4 module, because the energy released by the M4 module at the bottom pre-heats the M5 module, and the cooling plate makes the bottom heating more evenly. No obvious ordered propagation was observed in the M5 module but synchronous and disordered propagation. The double-layer structure also exists in the electric energy storage stations. Table 1 lists thermal runaway propagation (TRP) characteristics and trigger times involving M4 and M5.

The synchronous propagation and disordered propagation should be taken care in the battery system design, as most of the current researches are on preventing the ordered propagation. To effectively suppress both synchronous and disordered propagation in battery pack systems, various methods can be employed, such as adding materials with both heat absorption and insulation properties to the side of the double-row module, installing a device that directs and exhausts the hot gas produced inside the battery pack to the outside, designing the liquid cooling system as an immersion cooling method, and installing a pressurized fluorohexane fire suppression system.

3.1.3. Analysis of thermal runaway propagation sequence

The triggering of TR in LIBs is due to any position of the jelly roll inside the battery reaching T_2 . The essence of the propagation sequence of thermal runaway is the order in which the cells in

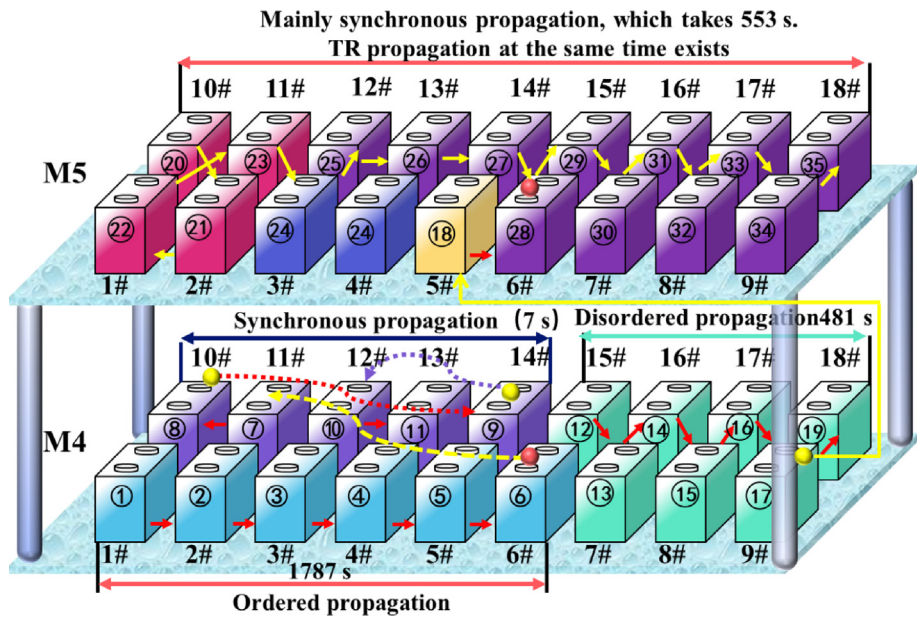


Fig. 5. The thermal runaway propagation sequence in a CTP battery system.

Table 1
TR time and characteristics about M4 and M5.

M4	Cell	1	2	3	4	5	6	7	8	9
	Time	797 s	1157 s	1512 s	1913 s	2254 s	2584 s	2788 s	3005 s	3135 s
	TRP	Ordered propagation					Disordered propagation			
	Cell	10	11	12	13	14	15	16	17	18
M5	Time	2678 s	2675 s	2681 s	2682 s	2680 s	2764 s	2944 s	3088 s	3245 s
	TRP	Synchronous and disordered propagation					Disordered propagation			
	Cell	1	2	3	4	5	6	7	8	9
	Time	3478 s	3475 s	3500 s	3500 s	3220 s	3593 s	3645 s	3705 s	3749 s
	TRP	Synchronous			Same moment		First TR		Disordered propagation	
	Cell	10	11	12	13	14	15	16	17	18
	Time	3467 s	3488 s	3526 s	3561 s	3570 s	3623 s	3653 s	3731 s	3779 s
	TRP	Synchronous		Synchronous and disordered propagation						

the module, under different heat transfer modes, reach T_2 in their respective jelly rolls.

See Fig. S3, the propagation sequence in M4 module occurs in an ordered manner, with cells 1 to 6 undergoing sequential propagation. The primary cause of ordered propagation is the temperature gradient between the individual cells within the module, which is established during heating by the heater. When cell 1 undergoes thermal runaway, the other cells (2#–6#) undergo TR in sequence, driven by heat transfer through the solid surface of the previous cell. The propagation sequence follows the direction of heat flow provided by the solid surface of the heater and cells. In summary, the reason for ordered propagation can be attributed to the temperature gradient generated by solid surface heat transfer between individual cells in the module.

Synchronous propagation primarily occurs when there is little temperature difference between cells in a module, and is often observed in modules subject to lateral heating (i.e., heat transfer through the side of the cells during TR) or direct heating of vent gases and particles. During the process of triggering thermal runaway in the 1#–6# cells (i.e., ordered propagation) of the M4 module, the heat generated from the thermal runaway is transferred to the 10#–14# cells through the side, resulting in a continuous increase in the jelly roll temperature of the 10#–14# cells under the effect of lateral heat flow. Additionally, the high-temperature smoke and particles generated by the thermal runaway of the cells inside the M4 module heat up the liquid cooling plate at the bot-

tom of the M5 module. The cells inside the M5 module are subjected to vertical heat flow and heating caused by the flow of high-temperature smoke, causing a continuous increase in the jelly roll temperature inside the cells.

The heat transfer mechanism of disordered propagation involves the coupling of heat transfer through a large surface area, lateral heat transfer, and heat transfer through vent gases. In addition, differences in the quality of the cells themselves can also affect the time when the cells trigger TR, even under the same heat transfer conditions. In the case of disordered propagation, it is unclear which type of heat transfer effect plays a critical role in triggering TR, as it is a complex interplay of various factors. Under such combined effects, TR will occur when the jelly roll temperature of any cell reaches T_2 .

Based on the aforementioned analysis, the primary cause of ordered, synchronous, and disordered propagation within the module is the different directions of heat flow entering the module and the cells. When heat flow mainly enters from the sides of the module and cells, the individual cell heating situation is generally uniform, resulting in relatively similar temperatures among each cell and making it more susceptible to synchronous propagation. Conversely, when the module is heated from the front or back, individual cells will absorb the majority of the heat and are more prone to TR, thus leading to sequential propagation within the module. The disordered propagation is a complex interplay of multiple heat transfer mechanisms and variations in cell quality.

3.2. Vent gas migration inside the Cell-to-Pack system

According to the data response for the sensors S1–S3, the smoke flow path in the CTP is $S2 \rightarrow S1 \rightarrow S3$. All the smoke sensors are destroyed after 2696 s, when 11 cells in the M4 module have triggered TR, as shown in the Fig. S4.

12 VOC detectors helped visualize the gas migration (especially the organic solvent) behavior inside the battery pack. Except for the gas vented during TR, the carbonate solvents will vaporize and vent out under boiling temperature. The cyan arrows depicted the migration paths of the carbonate solvent after bursting out. The diffusion order of the TR gas was: $V1(781\text{ s}) \rightarrow V5(806\text{ s}) \rightarrow V8(842\text{ s}) \rightarrow V6(855\text{ s}) \rightarrow V3(878\text{ s}) \rightarrow V7(895\text{ s}) \rightarrow V4(927\text{ s}) \rightarrow V10(983\text{ s}) \rightarrow V9(1103\text{ s}) \rightarrow V11(1158\text{ s}) \rightarrow V12(1400\text{ s})$.

The time when the 1# cell in M4 underwent TR is regarded as 0 s, then the gas diffusion to the 12# VOC sensor takes 1400 s. This moment corresponded to the moment when the #3 cell in M4 was triggered into TR. The steps of the VOC curve correspond to TR propagation in different cells. The pulse response of the VOC sensor is slightly faster than that of the smoke sensor. Using the smoke sensor or the VOC sensor as early warning signal might be reconsidered, because the location of the gas sensor will influence the detection sensitivity. Too far distance may cause no signal response recorded by the gas sensor.

3.3. Temperature response inside the CTP

The temperature characteristics at different positions in the CTP can provide data support for optimizing the safety design against failure propagation. The temperature at different places inside the CTP was collected, including the side temperature of M1, M2, and M3, the temperature of the liquid cooling plate between M4 and M5, the inside and outside temperature of the top cover, the temperature of the air domain surrounding the BMS, etc. Fig. S5 (a–c) shows that, M1, M2, and M3 do not suffer TR, even when the TR propagation of the M4 and M5 modules ended. The temperature at the side of the cells in M1, M2, and M3 shows that the temperature of the whole module is between 90 and 170 °C throughout the entire process of TR propagation. Spikes that refer to jet flame were not considered when we say mild temperature between 90 and 170 °C. The spikes indicate that the hot gas vented from M4 and M5 is squeezed into the sleeves between M1–M3 and the battery pack cover. The magnitude rise in temperature for the M1–M3 modules corresponded to the synchronous propagation and stable combustion of M4 and M5. However, as there is a 45.6 cm air gap that segregates the M1–M3 from M4 and M5, the temperature rise is not very high. The ambient temperature at the BMS region increased slightly from 100 to 400 °C, as shown in Fig. S5(d).

The liquid cooling plate under M5 blocked the initial heat radiation from M4 at the bottom. Fig. S5(f) illustrates the temperature profile of the sensors at the liquid cooling plate between M4 and M5. The cooling plate survived from the heat radiation at the stage of “ordered propagation” in M4, whereas its temperature rose in pulses when the “synchronous propagation” started in M4. The liquid cooling plate was melted once the TR in the M5 module started, because the temperature exceeded the melting point of the aluminum plate.

The temperature inside and outside the top cover of the battery pack can reflect the migration behavior of vent gas. Fig. S5(g and h) shows that before the occurrence of synchronous propagation, the highest temperature at the inner surface of the top cover is always lower than 300 °C, and after synchronous propagation, the temperature rises to between 300 and 700 °C. The highest temperature of the air domain in the battery pack is concentrated at the area above the BMS, which is marked by the red circle in Fig. S5(g). This

area, where the electrical device is installed, has a lifted shape that makes it easier to accumulate heat. It also blocks the hot gas migration from M4 and M5 to M1–M3. Fig. S5(i) shows the temperature curve adjacent to the VOC sensor, representing the temperature field in the internal space of the CTP. The pulse shape in the temperature curve indicates that the hot gas passes the VOC sensor like pulses. The temperature range was between 132 and 624 °C, with the highest temperature occurring when the flame was observed outside the CTP system at 3497 s.

3.4. Characteristics of smoke and fire

Fig. S6 shows time sequence of gas venting and flame jetting. The moment when the 1# cell in M4 underwent TR is defined as absolute zero time. The gas generated by the TR of the 1# cell caused a pressure increase inside the CTP system. The pressure relief valve of the CTP was opened when the pressure reached 20 kPa. This was the expected phenomenon of TR in CTP, because the controlled venting helped dissipate heat and avoid the accumulation of flammable gas. However, the top cover near the 1# cell was not strong enough to hold intact, and it was broken at 10 s after the 1# cell was triggered into TR. As a consequence, a large amount of black smoke was ejected outward from the fractured area near the 1# cell. At 1894 s, the synchronous TR propagation occurred, with the gas venting out violently, accompanied by sporadic red sparks. As the synchronous propagation went on, the morphology of the vent gas kept changing periodically. The black smoke and grey smoke occurred alternatively. The periodical changing stopped when a deflagration was observed at the fracture point. The precursor of the deflagration of the battery pack might be the appearance of periodic black gas and grey gas. As the deflagration causes severe damage to the surroundings, the firefighters are suggested to adjust their tactics when similar scenery was observed.

The key to the combustion of the gas generated during the TR of LIBs is to satisfy the fire tetrahedron: fuel, oxidizer, ignition source, and uninhibited chemical chain reaction. During the ordered propagation of the TR of the CTP battery pack, the damage to the top cover of the battery pack was relatively small, and the combustible gas generated during the TR of the battery did not reach the flammable limit after mixing with the oxygen inside the battery pack. In addition, the heat released during the ordered propagation was not enough to ignite the combustible gas, resulting in no visible flames during the ordered propagation process. During the synchronous propagation process, the TR energy released by the battery module was greater than that during the ordered propagation. The high-temperature smoke burned the top cover of the battery pack, causing a large amount of air to enter the interior of the battery pack and mix with the combustible gas to reach the flammable limit. The combustion and explosion occurred under the effect of high-temperature particles, electrostatic sparks, or high-temperature surfaces released during the TR.

Fig. S7 shows the typical combustion characteristics of the CTP after the appearance of the flame. The start of jet flame significantly accelerated the failure propagation in the CTP system. The ejection flame shot out along the edges of the battery pack, followed by a spherical flame striking out. When the synchronous propagation started in M5, the TR behavior was even more violent, and the jelly rolls inside the battery cells were throwing out with the jetted flame.

The direction of flame jetting was switched from side direction to upward direction when the synchronous propagation occurred in M5. The temperature of the vent gas and jet flame was recorded by infrared thermometers. The battery pack temperature was between 17.5 and 232.7 °C, when there was no flame. The temperature instantly rose to approximately 400 °C at the moment of

ignition. The highest recorded temperature of the battery pack was up to 1100 °C during the phase of violent ejection. The battery pack temperature gradually dropped below 600 °C after the TR propagation was over.

3.5. Mechanical damage features

The mechanical damage of the CTP system includes the mass loss and deformation of battery cells, providing references for accident investigation. Fig. 6 concludes the mass loss of the battery cells in M4 and M5.

Among them, Fig. 6(a and c) is drawn according to the battery number, and Fig. 6(b and d) is drawn according to the battery failure sequence. See Fig. 6(b and d), the mass loss caused by the synchronous propagation is more consistent, i.e., the mass loss would be more stable when the TR propagation time is shorter. For instance, the mass loss rate of the cells in the M4 module was between 31% and 59%, which fluctuated greatly, as shown in the Fig. 6(e). The mass loss rate of the cells in the M5 module was between 46% and 56%, as shown in the Fig. 6(f). Intensive combustion occurred in the latest stage of TR propagation of the CTP battery pack. In the CTP, the TR propagation behavior under the synergistic coupling of “hot vent gas, flame, and solid heat transfer” will cause more significant mass loss. For instance, the cells 11#–14# in the M4 module that underwent synchronous propagation had 55% or more mass loss. In addition, the extent of damage

to the battery case will be more serious during synchronous propagation.

Afterwards, we disassembled the modules after TR test to explore the deformation characteristics of the battery cells. Fig. S8 shows the features of the debris of M4 and M5 after TR. The damage of M5 after TR is more serious than that of M4. The damage degree of the shell for the battery that underwent TR later was more severe than that for the battery that underwent TR earlier. This is because, in the process of TR propagation, the battery that underwent TR later absorbed more energy due to the preheating from the previous failed battery. During the process of triggering TR, the internal materials of the battery react and release gas, causing the battery to expand and squeeze adjacent cells. The surface of the non-TR battery is relatively hard, and the expansion stress of the TR battery can only be released towards the remnants of the TR battery, causing a depression on the surface of the TR battery remnants. Therefore, during TR propagation in the battery module, the direction of the protrusion of the TR battery remnants is opposite to the direction of the heat flow during TR. If the battery that first undergoes TR is located in the middle of the module, it will exhibit the characteristic of a double-sided concave shape. According to the bulge directions marked in Fig. S8, the first cell that underwent TR is most likely A1 (highlighted in red) in the figure. The characteristic that the protrusion direction of the battery remnants is opposite to the direction of heat flow during thermal runaway propagation is a commonly used method for determining the first battery in the battery module that triggered TR.

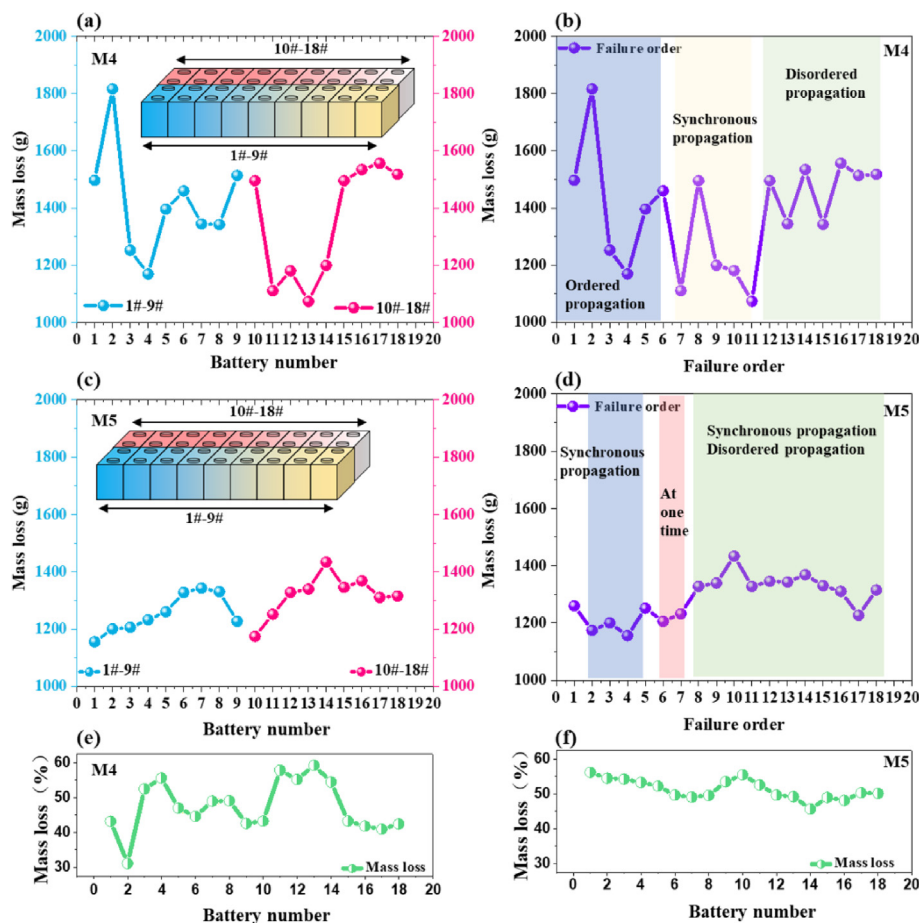


Fig. 6. The mass loss of the cell after thermal runaway propagation in the CTP system. (a) Mass loss of the battery in M4 by spatial position; (b) mass loss of the battery in M4 by failure order position; (c) mass loss of the battery in M5 by spatial position; (d) mass loss of the battery in M5 by failure order position; (e) mass loss percentage of the battery in M4 by space position; (f) mass loss percentage of the battery in M5 by space position.

3.6. Physical and chemical characteristics of smoke particles

SEM+EDS, XRD, ICP-MS, XPS, and CHNSO were used to characterize the composition of the smoke particles. The smoke particles are mainly from the TR battery and deposited on the non-TR battery modules. Fig. 7 shows the results for physical and chemical analysis.

The microstructure of the smoke particles shows different granularity and smoothness, as shown in Fig. 7(a), and the elemental analysis by EDS indicates that the particles mostly consist of O, C, S, F, Co, Cu, Mn, Ni, and P. Quantitative CHNSO and ICP-MS tests were carried out to determine all the elements that were contained in the smoke particles, as shown in Fig. 7(b). The smoke particle contains 22.2% O, 0.09% H, 22.1% C, as collected in smaller bar chart. The bar chart shows the actual content of some elements obtained by the ICP-MS test with a unit of mg/kg, as shown in Fig. 7(b). The XRD results show that the smoke particles mainly contain C, NiO, LiF, Ni/Co metal, and MnO, as shown in Fig. 7(c). The microscopic morphology indicates that the particles with relatively smooth surfaces mainly contain carbon. The particles with a rather rough surface are more likely to be metals and metal oxides, and the flocs on the surface of particles mainly contain fluoride. Ref. [25] reported that there is Li_2CO_3 in the vented particle. However, there was no Li_2CO_3 observed in the smoke particles in this research. The smoke particles are light, and the particle size is generally less than 100 μm . The small size makes it easy to adhere to the cold surface of objects. The Li_2CO_3 can be detected in the battery residual jelly roll, where the particulate matter usually has larger particle size ($\geq 100 \mu\text{m}$) after TR occurred.

XPS is a technique for analyzing the chemical properties on the surface of a substance [33]. The XPS can help characterize the elemental composition, chemical state, and electronic state in materials. The valence and chemical bond of element can be further obtained by fitting the characteristic peaks. Fig. 8(a) shows the full spectrum of the entire sample, in which the orbits of each element

are marked. Fig. 8(b) shows the peak-splitting fitting results of the 2p orbital for the Al element. There is a peak located at 74.9 eV, indicating that the Al element in the material exists in the form of Al^{+3} .

Fig. 8(c) shows the peak-splitting fitting diagram of the 1s orbital for the carbon element, which contains the bond of C–C (284.6 eV), C–O (285.4 eV), C=O (286.3 eV), C–F (289.7 eV) and other groups. This is consistent with the peak-splitting fitting results of the F 1s orbits in Fig. 8(d) and the O 1s orbits in Fig. 8(e). The Mn and Ni elements in the sample exist in the form of two valence states. Fig. 8(f) shows the peak fitting diagram of the Mn 2p orbit. The two orbitals of Mn 2p can be divided into two valence states, i.e., Mn^{+3} and Mn^{+4} . There are four characteristic peaks of the Mn element, among which the peaks at 641.7 and 644.6 eV are the characteristic peaks of Mn 2p_{2/3} orbits, belonging to Mn^{+3} and Mn^{+4} species, respectively [34,35]. According to Fig. 8(f), the peaks at 653.0 and 654.4 eV are the Mn 2p_{1/2} orbits belonging to Mn^{+4} and Mn^{+3} species, respectively. The Ni element contains six characteristic peaks, as shown in Fig. 8(g). Among them, the peaks at 855.5 and 856.9 eV are the characteristic peaks of the orbit Ni 2p_{3/2}. They belong to species Ni^{+3} and Ni^{+4} , respectively. The peaks at 873.3 and 875.1 eV are characteristic peaks of orbit Ni 2p_{1/2}, belonging to Ni^{+2} and Ni^{+3} , respectively. In addition, the peaks at 861.7 and 875.1 eV are satellite peaks of Ni^{+2} and Ni^{+3} , respectively. From the valence law of elements Ni and Mn, it can be inferred that there was violent redox reaction in the battery cell that suffered TR propagation.

See Fig. 8(h), the peaks of P 2p_{1/2} and P 2p_{3/2} at 134.5 eV and 133.6 eV indicate the presence of the P element in the sample. The peak near 169.4 eV is attributed to the S 2p orbital, but the content is relatively small, which is caused by the low content of S in the sample, as shown in Fig. 8(i). For the 1s orbital of Li element, there is a peak attributed to LiF, with the peak center located at 56.0 eV, which corresponds to the results of the peak fitting of the F 1s orbital, corresponding to LiF (685.6 eV), as shown in

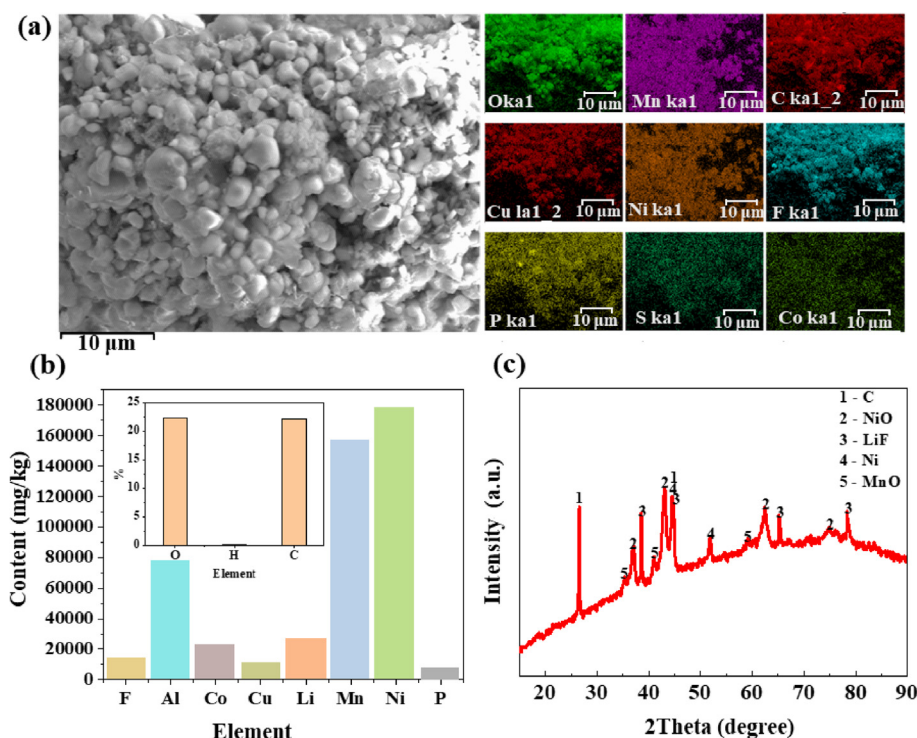


Fig. 7. (a) SEM image and EDS analysis; (b) quantitative result by CHNSO and ICP-MS; (c) XRD result.

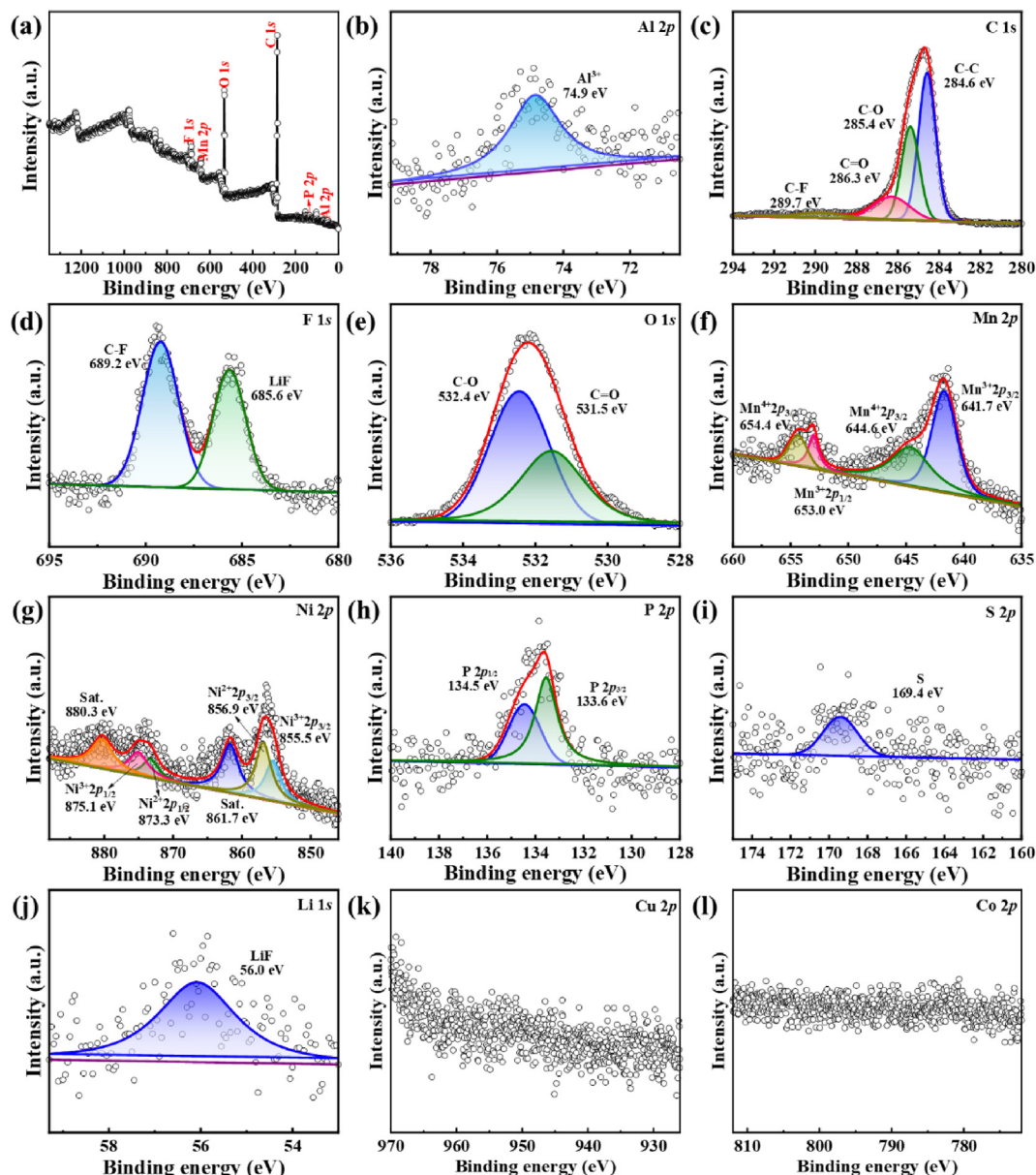


Fig. 8. XPS spectra. (a) Full spectrum; (b) Al 2p; (c) C 1s; (d) F 1s; (e) O 1s; (f) Mn 2p; (g) Ni 2p; (h) P 2p; (i) S 2p; (j) Li 1s; (k) Cu 2p; (l) Co 2p.

Fig. 8(j). As shown in Fig. 8(k and l), for the orbitals of Co 2p and Cu 2p, there is no pronounced peak, indicating the low content of Co and Cu in the residual.

The chemical analysis of smoke particles can guide the investigation of battery TR accidents. Accident investigators can trace the cause by analyzing the physical and chemical characteristics of the smoke particles after battery TR. The candidate analysis includes the microscopic morphology, element content, compound composition, and the change of the element valence. The peak strength and peak area of elements in the smoke particles released by TR of LIBs are different, such as Al, C, F, O, Mn, Ni, and P, and these elements are called the sensitive elements. On the contrary, the content and peak intensity of Co, Cu, S, and Li elements in the analysis are low, called the insensitive elements. The analysis of sensitive element can be carried out on the smoke particles extracted from the failure scene during the investigation of the TR accident of LIB. The XPS analysis results of those insensitive elements can be ignored, thereby improving the efficiency of investigating the causes of TR for LIBs.

4. Conclusions

This research studied the thermal runaway propagation characteristics of a Cell-to-Pack battery system. In addition, the characteristics of temperature response, the behavior of vent gas and jet flame, the mass loss and deformation of the battery, and the physical and chemical properties of smoke particles were studied by quantitative analysis methods, leading to the following conclusions.

- (1) The thermal runaway propagation for the Cell-to-Pack battery system has three major patterns, including the ordered propagation, the synchronous propagation, and the disordered propagation. The synchronous propagation causes more severe damage on the batteries than the others.
- (2) The liquid cooling plate between the upper and lower modules cannot block thermal runaway propagation between modules. While the large gap where the battery manage-

ment system was placed helps prevent thermal runaway propagation from module to module.

- (3) The bulge direction points to the first TR battery in the ordered propagation process, and the synchronous propagation back leads to more severe damage and mass loss of the battery in the CTP.
- (4) In the TR propagation of CTP, if the temperature surge caused by flame is not taken into account, the temperature in the area where TR is not triggered is between 90 and 170 °C. It takes 619 s for the hot gas to reach the farthest end in the CTP from the triggering area. The time for the flame to appear outside the CTP battery pack is 45 min after the first battery triggers the TR.
- (5) According to the sensitivity of the change of valence states of the elements from the smoke deposit, the elements of the battery component materials are classified, among which the sensitive elements are Al, C, F, Mn, Ni, and P, whereas the insensitive elements are Co, Cu, S, Li, and P.

The Cell-to-Pack enables higher energy density for battery system through structural innovations. The fail-safe design of the Cell-to-Pack system is essential to ensure the safety during vehicle usage. Future research will focus on modeling the propagation behavior using multiphysics simulations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jechem.2023.05.015>.

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